

Impact of Marginal Extra Rest on Performance in the National Football League

Conducted by: Nolan Spenker, Treyvin Bundy, Olivia Gleason, Logan Talin

Executive Summary

1. Reason for the Project

National Football League (NFL) scheduling creates unavoidable rest day gaps between opponents, and Thursday Night Football (TNF) is the most consistent example. Teams playing on Thursday receive three to four additional days of recovery before their next game. This project examined whether that marginal rest advantage translates into measurable defensive improvements and identified which defensive position groups and metrics are most affected.

Understanding the impact of rest can help teams optimize game planning, practice schedules, roster decisions, and opponent preparation. Offensive and defensive coaches can use these findings to better account for rest differentials and potentially gain a competitive advantage. This project will focus on the defensive side based on the intuition that defenses are impacted more from extra rest days.

2. Quick overview of the Process to Answer the Questions

Game schedule data, player rosters, and play-by-play data were combined to calculate rest differentials between opponents. Matchups involving a team coming off a Thursday Night Football appearance were isolated to measure the effects of additional recovery time. Defensive performance was evaluated across defensive linemen, linebackers, defensive backs, and overall defense using position-specific metrics. Correlation analysis and regression were applied to determine whether rest differences produced significant and consistent effects on defensive outcomes.

3. Summary of Findings and Recommendations

Additional rest produced little measurable impact across most defensive metrics. Linebackers and defensive backs showed no meaningful improvements from extra recovery time. Overall defensive performance improved slightly with additional rest, though the practical effect was minimal. The strongest findings occurred along the defensive line, where short-rest defenses generated pressure at higher rates than expected.

It is recommended to place a higher emphasis on maintaining aggressive defensive line pressure packages for play-calling in short weeks. Defensive linemen have proven to generate effective pressure even with little recovery time. Coaches should not expect meaningful differences from linebackers and secondary positions from differing preparational timeframes. Rest days seem to provide marginal defensive advantages, if any.

Detailed Sections

1. Description and Discussion of the Development of the Analytics Question(s)

Rest differentials (the difference in recovery days between opponents) can influence player recovery, preparation time, and game performance. The clearest example is when a team plays on Thursday Night Football and receives additional days of rest before its next game, also known as a mini-bye.

Defense was selected as the primary area of focus because success depends heavily on physical responsiveness, stamina, and rapid decision-making; all of which may be influenced by fatigue and recovery. The central question guiding this analysis was:

How does the marginal rest advantage gained from a Thursday Night Football appearance impact defensive performance in a team's next game relative to its opponent?

The objective was to determine whether additional rest produces statistically significant improvements in defensive outcomes and to identify which position groups are most affected.

2. Discussion of Variables Available and Variables Needed to Examine the Analytics Question

Game schedules, player rosters, and play-by-play data from 2012 through 2024 (years of consistent TNF scheduling) were combined to evaluate how rest impacts defensive performance. Rest advantage was measured using the number of days between current games and previous games for each opponent, taking the difference between values to isolate disproportionate rest advantages.

Defensive performance was measured using points allowed (total points given up by the defense in a game), defensive expected points added (EPA, measure play-by-play value of the defense), success rate allowed (indicates whether the offense achieved a positive outcome on a given play), defensive line metrics (pass-rush effectiveness), linebacker metrics (run stopping and tackle statistics), defensive back metrics (primarily coverage statistics and indicators), and overall defensive metrics.

3. If you Constructed New Variables, Detail the Construction Method

- Extra Rest Days: Team rest days minus opponent rest days.
- Defensive EPA: Offensive EPA multiplied by -1 to invert for defensive effectiveness.
- Pressure and Stuff Indicators: Binary variables identifying quarterback pressures and unsuccessful rushing attempts. Pressure indicator was constructed through given pressure variables, and stuff indicator constructed using features showcasing play type and yards gained.
- Sack Yards: Built using given variables of play type and yards gained. If a sack was recorded, yardage indicated is taken.
- Position Group Variables: Created using player identifiers and roster information to attribute plays and statistics to defensive linemen, linebackers, or defensive backs.
- Shared Statistic Position Indicator: Notes which position group is responsible for shared defensive statistics, constructed through corresponding player ID in play-by-play data.

4. Flowchart of the Analytics Logic to Answer the Analytics Question(s)

See Appendix C for project flowchart.

5. Rationale for Utilizing the Analytics Tool(s) (Excel, Tableau, R, Python, Other)

Python was used for data cleaning, merging, and feature engineering. After preparation, Python, R, and Excel were all used for analysis, including statistical testing and visualization. Tool usage varied by team member to cater to individual strengths, but methods remained consistent.

6. Description of Conversion from Analytics Logic to Analytics Tool(s)

The analysis was divided by position group (see Appendix C), allowing each team member to focus on a specific component of the project using their preferred tool. This approach improved efficiency while maintaining a consistent methodology across all analyses. Findings from each section were combined to provide a comprehensive assessment.

7. Primary Findings

Defensive Line (DL)

Defenses on fewer than seven days of rest pressured quarterbacks on 22.17% of pass plays compared to 19.17% on standard rest. This effect was largest on first down, with short-rest defenses generating more pressure. Both of these differences were found to be statistically significant.

Higher pressure did not lead to more sacks or quarterback hits. Forced-fumble rate fell under short rest, indicating weaker finishing despite increased disruption. The relationship between pressure rate and defensive EPA strengthened under short rest, with higher pressure rate marginally correlated with higher defensive EPA.

Results suggest defensive coordinators should maintain aggressive pass-rush schemes during short weeks rather than reducing pressure.

Linebackers (LB)

No meaningful impact was observed for linebacker performance, with all LB-related metrics (tackles for loss, sacks, and interceptions) nearly identical for rested vs. non-rested groups. Defensive EPA was slightly higher for LBs with a mini-by (0.0168 compared to 0.0015) but was not statistically significant.

Defensive Backs (DB)

Results reveal no statistically significant impact across all key coverage metrics. Performance differences were minor and inconsistent, with high variability indicating that normal game-to-game fluctuations outweigh any impact from rest advantage. Regression analysis reveals extra rest explains less than 0.3% of variation in DB performance.

Overall Defense

Defensive EPA was used as a unified measure of defensive performance because it captures expected scoring impact on a play-by-play basis:

$$\text{EPA} = \text{Expected Points After Play} - \text{Expected Points Before Play}$$

A linear regression using defensive EPA and rest differential found a statistically significant relationship ($p = 0.03$), with each additional rest day increasing defensive EPA by approximately 0.007 per play. With 50,000+ observations, our R-squared value indicates that extra rest days explain little of the variation in defensive EPA. This suggests rest isn't a meaningful predictor despite the evidence.

Teams with a mini-by allowed 22.96 points on average compared to 23.64 for teams without one, a statistically significant but small practical difference.

8. Identification of Extensions & Recommendation to Accomplish the Extensions

High-impact in-game situations and whether extra preparation creates an advantage: Extension of the analysis to high-leverage situations such as third downs and red zone defense. Existing play-by-play data can support initial comparisons of third-down efficiency and red zone defensive performance under different rest conditions.

Offensive production: Extension of the framework to the offensive side of the ball to determine whether additional rest improves scoring efficiency. The same dataset can be used to evaluate points scored, third-down conversion rate, and red zone touchdown rate. Incorporating offensive line grading or pressure data would strengthen the analysis of preparation effects.

Appendices

Appendix A: Data File

The dataset used for analysis can be found [at this link](#).

Appendix B: Code/Analysis File

The code and analysis used to inform this report can be found [at this link](#).

Appendix C: Flow Chart

